MIL-F-8115C(USAF)
21 December 1960

Superseding
MIL-F-8115B
17 November 1958

MILITARY SPECIFICATION

FLOODLIGHT, INCANDESCENT, EXPLOSION PROOF, TYPE MA-1

1. SCOPE

1.1 This specification covers one type of explosion-proof, incandescent floodlight, designated MA-1.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Federal

J-C-90	Cable, Power, Electrical (Flexible Cord And Fixture Wire)

Military

MIL-P-116	Preservation, Methods of
MIL-P-7936	Parts and Equipment, Aeronautical, Preparation For Delivery
MIL-C-7989	Covers, Light-Transmitting, For Aeronautical Lights
MIL-D-70327	Drawings, Engineering and Associated Lists

STANDARD

Military

MIL-STD-129	Marking for Shipment and Storage
MIL-STD-130	Identification Marking of U.S. Military Property
MS33586	Metals, Definition of Dissimilar

FSC 6230

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PUBLICATION

Air Force-Navy Aeronautical Bulletin

143 Specifications and Standard; Use of

(Copies of documents required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

2.2 Other Publication.- The following document forms a part of this specification. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

UNDERWRITERS' LABORATORIES, INCORPORATED

List of Inspected Electrical Equipment

Standard 844 Fixtures, Electric Lighting, For Use in Hazardous Locations

(Copies of Underwriters' Laboratories' publications may be obtained from Underwriters' Laboratories', Incorporated, 161 Sixth Avenue, New York 13, New York; 207 East Ohio Street, Chicago, Illinois; or 1655 Scott Lane, Santa Clara, California.)

3. REQUIREMENTS

3.1 General Specification.- The requirements of Underwriters' Laboratories Incorporated apply as requirements of this specification. When the two specifications conflict, this specification shall govern.

3.2 Components.- The floodlight shall consist of the following:

Item No.	Quantity	Description	See Requirements
1	1	Fixture assembly	3.7
2	1	Base	3.8
3	1	Power cable	3.9

3.3 Selection of Specifications and Standards.- Specifications and standards for necessary commodities and services not specified herein shall be selected in accordance with Bulletin 143 except as provided in 3.3.1 and 3.3.2.

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3.3.1 Commercial Parts.- Commercial parts having suitable properties may be used where, on the date of invitation for bids, there are no standard parts. In any case, commercial utility parts such as screws, bolts, nuts, and cotter pins, having suitable properties may be used provided:

a. They can be replaced by the standard parts (MS or AN) without alteration.

b. The corresponding standard part numbers are referenced in the parts list and, if practicable, on the contractor's drawings.

3.3.2 Standard Parts.- With the exception in 3.3.1, MS and AN standard parts shall be used where they suit the purpose. They shall be identified on the drawings by their part numbers.

3.4 Materials

3.4.1 Fungus-Proof Materials.- Materials that are nutrients for fungi shall not be used where it is practical to avoid them. Where used and not hermetically sealed, they shall be treated with a fungicidal agent acceptable to the procuring activity. However, if they will be used in a hermetically sealed inclosure, fungicidal treatment will not be necessary.

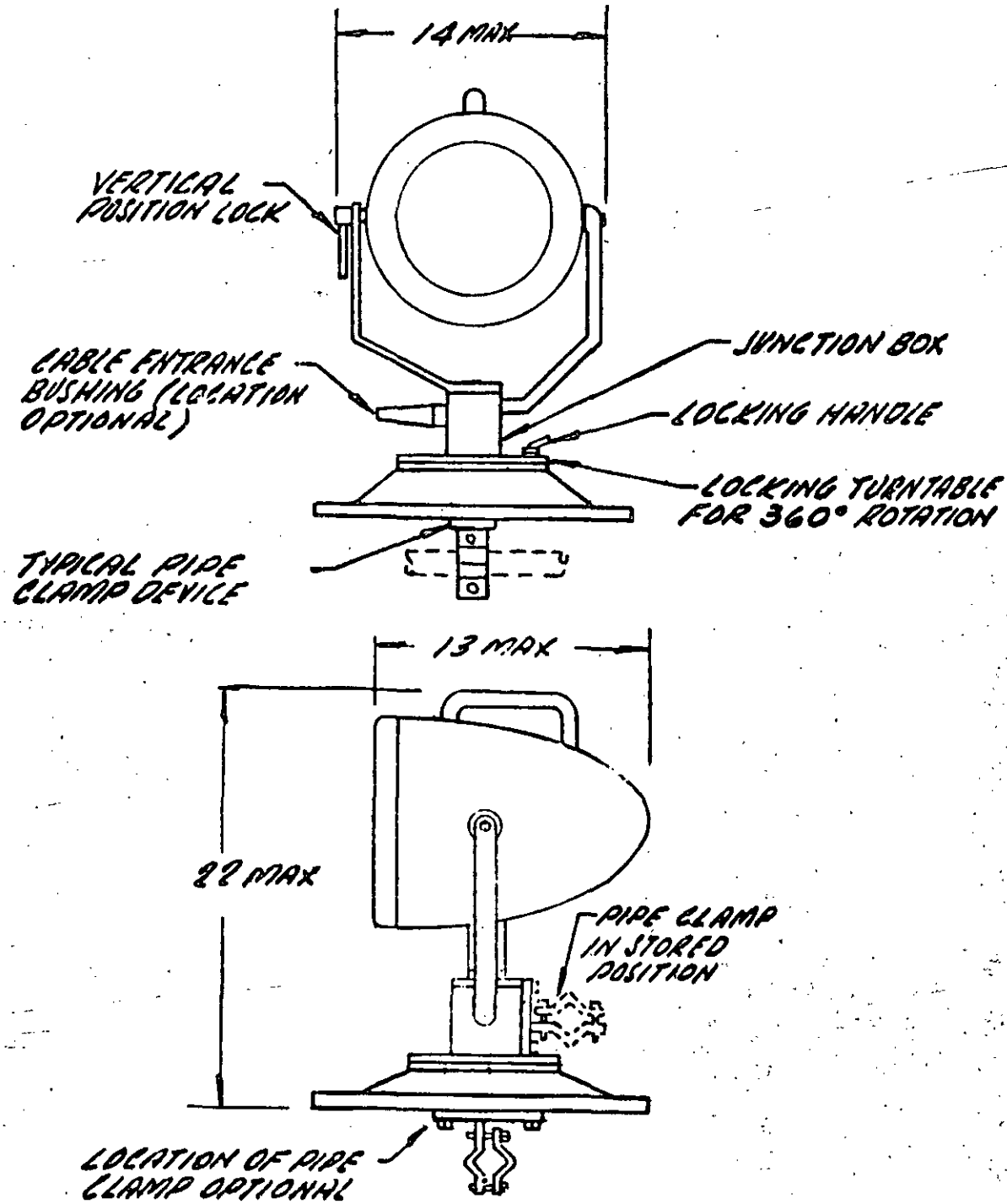
3.4.2 Metals.- Metals shall be of the corrosion-resistant type or treated to resist corrosion due to fuels, salt spray, or atmospheric conditions as may be encountered in storage or normal service.

3.4.2.1 Dissimilar Metals.- Unless suitably protected against electrolytic corrosion, dissimilar metals shall not be used in intimate contact with each other. Dissimilar metals are defined in MS33586.

3.5 Design.- The design of the floodlight shall be essentially as shown on figure 1. The figure shown is a representative sample, and assemblies of slightly different design will be considered as meeting the requirements provided the specified dimensions and all other requirements are met.

3.6 Reliable Construction.- The floodlight shall be so constructed that no parts will work loose in service. It shall be built to withstand the strains, jars, vibrations, and other conditions incident to shipping, storage, installation, and service. All exposed metal parts shall be cast or fabricated from nonsparking metal.

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DIMENSIONS IN INCHES UNLESS OTHERWISE SPECIFIED - TOLERANCE: FRACTIONS $\pm \frac{1}{32}$

FIGURE 1. FLOODLIGHT.

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3.6.1 Adjustments and Repairs.- The floodlight shall be so constructed that lamp replacements may be easily made without the use of special tools, and all other adjustments and repairs may be easily made by the personnel of operating units and overhaul bases.

3.7 Fixture Assembly.- Each fixture assembly shall include a lens assembly, carrying handle, yoke, turntable, clamp, and socket. A lamp support shall be provided to prevent lamp breakage during handling. The support shall not interfere with lamp replacement.

3.7.1 Lens Assembly.- The lens assembly shall be designed for removal without the aid of special tools. The glass used in the lens shall be class B, heat-resistant, clear glass in accordance with MIL-G-7989. The location and size shall be such that it will not interfere with the main beam of the lamp installed in the floodlight.

3.7.2 Carrying Handle.- A carrying handle shall be provided.

3.7.3 Mounting.- The flexibility of positioning as required by 3.7.3.1 and 3.7.3.2 shall be provided for both the normal base mounting and the clamp mounting.

3.7.3.1 Yoke.- The floodlight shall be provided with a yoke for holding the fixture assembly. The design of the yoke shall be such that it will allow the light beam to be elevated not less than 90° above, and lowered not less than 45° below the horizontal when the base is horizontal. A means shall be provided for securely fastening the floodlight in any position in the specified vertical plane of adjustment.

3.7.3.2 Turntable.- The light shall have a provision for positioning and securely clamping the floodlight through 360° of rotation in the horizontal plane (parallel to the base). This shall be accomplished by means of a locking turntable or similar device.

3.7.4 Clamp.- The basic design of the floodlight shall include a clamp for securely fastening the floodlight to vertical or horizontal tubing ranging from 1/2 inch to 2-3/16 inches in outside diameter. The clamp may be utilized as a means of fastening the lens assembly to the base. Otherwise, the clamp may be similar to the device shown on figure 1. A means shall be provided for storing the clamp on the light when the clamp is not in use.

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3.7.5 Socket.- The lamp socket shall be of a design that will accomodate any one of the lamps specified in 3.10. The socket shall be constructed of porcelain or other high temperature resistant materials having a minimum rating of 660w, 250V. The threaded socket shell shall be reinforced to prevent deformation if the lamp is screwed in too tightly.

3.8 Base.- The base of the floodlight shall be ample to securely support the floodlight when placed upon a floor or other flat surface, and to prevent accidental tipping. The base shall be similar to the base shown on figure 1.

3.9 Power Cable.- The floodlight shall be equipped with 100 ± 1 feet of No. 16, 3-conductor, type V or V1, flexible cord in accordance with J-C-90. One conductor shall be grounded to the floodlight for connection to the ground terminal of the plug. The power cable shall be supplied less the connector plug.

3.10 Lamps.- The floodlight shall accomodate any of the following lamps which shall not be furnished with the assembly.

a. A 300w, 120V R-40, medium-screw base, reflector floodlamp.

b. A 150w, 120V, PAR-38, medium-skirted base, projector floodlamp.

c. A 75w, 30V, PAR-38 medium-skirted base projector floodlamp.

3.11 The floodlight shall conform to all requirements of the Underwriters' Laboratories' Standard 844 for use in class I, group D, hazardous locations.

3.12 Part Numbering of Interchangeable Parts.- All parts having the same manufacturer's part number shall be functionally and dimensionally interchangeable. The item identification and part number requirements of MIL-D-70327 shall govern manufacturer's part numbers and changes thereto.

3.13 Dimensions.- The floodlight shall meet the dimensional requirements specified on figure 1.

3.14 Weight.- The weight of the floodlight complete shall not be more than 35 pounds.

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3.15 Operation Markings

3.15.1 Warning Plate.- Each floodlight shall have the following information molded into it, or provided on a plate conveniently and conspicuously located:

CAUTION

If the connection of the electrical circuit is in an area where explosive gases are present, an explosion-proof plug and receptacle shall be used. If the electrical connection is made outside of the hazardous area, a three-way grounded plug may be used.

3.15.2 Lamp Replacement Warning.- The following information shall be molded into the light, provided on a separate plate securely attached in a conspicuous place or included on the warning plate specified in 3.15.1:

Do not replace the lamp or disassembly floodlight in a hazardous location. Use one of the following lamps depending on service voltage and the required wattage:

300w, 120V, R-40
150w, 120V, PAR-38
75w, 30V, PAR-38

3.16 Identification of Product.- Equipment, assemblies, and parts shall be marked for identification in accordance with MIL-STD-130. The following additional information shall be included:

300w, 120V, Max.

3.16.1 Label.- Each lamp assembly shall bear the label of the Underwriters' Laboratories, Incorporated for use in class I, group D, hazardous locations.

3.17 Workmanship.- The floodlight, including all parts and accessories, shall be fabricated and finished in accordance with the highest quality commercial standards.

3.17.1 Riveting.- Riveting operations shall be carefully performed to insure tightness and satisfactory heading of rivets.

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3.17.2 Cleaning.- The floodlight shall be thoroughly cleaned, and loose, spattered, or excess solder, metal chips, and other foreign material removed during and after final assembly.

3.18 Approval.- The floodlight and cable shall be approved by the Underwriters' Laboratories, Incorporated as an electric lighting fixture suitable for use in Class I, Group D hazardous locations. (See 6.2.1)

4. QUALITY ASSURANCE PROVISIONS

4.1 Unless otherwise specified herein the supplier is responsible for the performance of all inspection requirements prior to submission for Government inspection and acceptance. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. Inspection records of the examination and tests shall be kept complete and available to the Government as specified in the contract or order.

4.2 Acceptance Tests.- Acceptance tests shall consist of the individual tests.

4.2.1 Individual Tests.- Each floodlight shall be subjected to the following tests, as described under 4.3:

a. Examination of product

b. Operation

4.3 Test Methods

4.3.1 Examination of Product.- Each floodlight shall be inspected to determine compliance with the requirements specified herein with respect to materials, workmanship, dimensions, and marking.

4.3.1.1 The floodlight shall be examined for proper operation of all parts and for compliance with all requirements specified in the applicable Underwriters' Laboratories, Inc., standards.

4.3.2 Operation.- The floodlight shall be completely assembled, equipped with lamp specified, and operated. It shall be inspected to determine proper operation of all adjustments and parts.

5. PREPARATION FOR DELIVERY

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5.1 General.- The floodlight shall be prepared for delivery in accordance with MIL-P-7936. For level A shipment, each floodlight shall be preserved in accordance with method IA-16 of MIL-P-116. The unit quantity shall be one each.

5.2 Marking.- Interior and exterior containers shall be marked in accordance with Standard MIL-STD-129. The nomenclature shall be as follows:

Floodlight, Incandescent, Explosion Proof Type MA-1

6. NOTES

6.1 Intended Use.- The Type MA-1 floodlight covered by this specification is intended for use as a worklight in areas where explosive gases may be present. For mounting, the light may be used with the wheel base provided, or clamped to the tubing members of aircraft maintenance stands. The pipe-clamp feature (see figure 1) is intended to afford convenient means for temporarily mounting the light adjacent to the working area. This feature is not to be construed as providing means for permanently installing the light.

6.2 Ordering Data.- Procurement documents should specify the following:

a. Title, number, and date of this specification
b. Required levels of preservation, packaging, and packing (see section 5).

6.2.1 Electrical Hazards Certification.- The bidder shall submit proof to the procuring activity that the floodlight and cable which he proposes to supply under this specification is approved by the Underwriters' Laboratories, Inc., as specified in 3.18. The label or listing of Underwriters' Laboratories, Inc., will be accepted as evidence of approval.

Notice. When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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